

Solution:

1.

$$y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x}$$

$$\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$$

$$= \frac{1}{\left(\frac{dy}{dx}\right)} \cdot \frac{dy}{dt}$$

$$\text{At } x = \ln 3, \frac{dx}{dt} = \frac{1}{2e^{2\ln 3}} (3) = \frac{1}{6}$$

2.

$$4^2 - 4(k)(2k - 7) < 0$$

$$16 - 8k^2 + 28k < 0$$

$$-2k^2 + 7k + 4 < 0$$

$$(2k + 1)(-k + 4) < 0$$

$$k < -\frac{1}{2} \text{ or } k > 4$$

However $k > 0$, therefore answer is $k > 4$

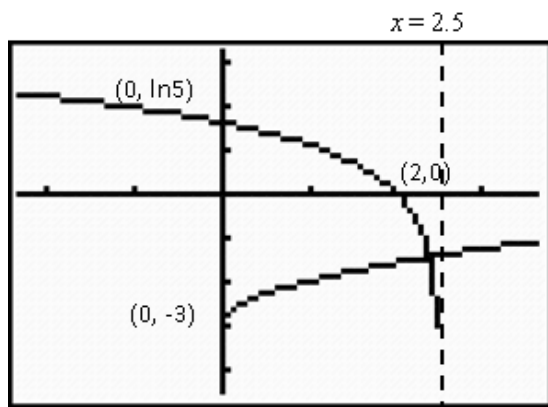
3.

$$(i) \int \frac{1}{\sqrt{1-2x}} dx = \frac{(1-2x)^{1/2}}{1/2(-2)} + C = -\sqrt{1-2x} + C$$

$$(ii) \int_3^8 \frac{1}{1-3x} dx = \left[\frac{1}{-3} \ln|1-3x| \right]_3^8 = \frac{1}{-3} (\ln 23 - \ln 8) = \frac{1}{3} \ln \left(\frac{8}{23} \right)$$

4.

(i)



(ii) $(2.3834, -1.4562)$

(iii) $\int_0^{2.3834} \ln(5-2x) - (\sqrt{x} - 3) dx = 6.507$

5.

$$y = \frac{8}{3}(2x-1)^{-3} + \ln(2x-1)$$

$$\frac{dy}{dx} = -8(2x-1)^{-4}(2) + \frac{2}{2x-1}$$

Let $\frac{dy}{dx} = 0$

$$\frac{-16}{(2x-1)^4} + \frac{2}{2x-1} = 0$$

$$\frac{-16 + 2(2x-1)^3}{(2x-1)^4} = 0$$

$$(2x-1)^3 = 8$$

$$2x-1 = 2$$

$$x = 1.5$$

Therefore, there is only one stationary point.

(ii)

At point P, $x=1, y = \frac{8}{3}, \frac{dy}{dx} = \frac{-16}{1} + \frac{2}{1} = -14$

$$\frac{y - \frac{8}{3}}{x-1} = -14$$

$$y = -14x + \frac{50}{3}$$

(iii)

Point Q is $\left(0, \frac{50}{3}\right)$ Area of triangle OPQ = $\frac{1}{2} \left(\frac{50}{3}\right)(1) = \frac{25}{3}$

6

(i)

$$\text{Vol} = \frac{1}{2} x^2 \frac{\sqrt{3}}{2} h = 1000$$

$$\therefore h = \frac{4000}{\sqrt{3}x^2}$$

$$\begin{aligned} \text{Surface area, } S &= 2 \left[\frac{1}{2} x^2 \frac{\sqrt{3}}{2} \right] + 3xh \\ &= 2 \left[\frac{1}{2} x^2 \frac{\sqrt{3}}{2} \right] + 3x \frac{4000}{\sqrt{3}x^2} \\ &= \frac{\sqrt{3}}{2} x^2 + \frac{4000\sqrt{3}}{x} \dots (1) \end{aligned}$$

(ii)

$$\frac{dS}{dx} = \frac{\sqrt{3}}{2} (2x) - \frac{4000\sqrt{3}}{x^2} = 0$$

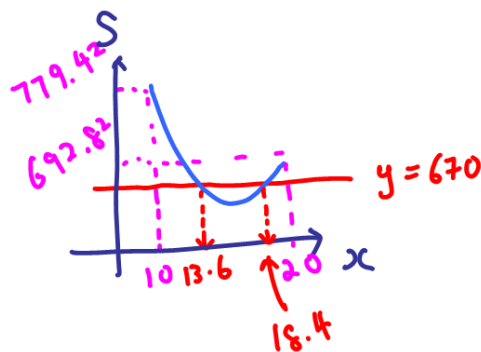
$$\Rightarrow \sqrt{3}x = \frac{4000\sqrt{3}}{x^2}$$

$$\Rightarrow x^3 = 4000$$

$$\Rightarrow x = 4000^{\frac{1}{3}}$$

Substitute $x = 4000^{\frac{1}{3}}$ into equation (1), $S = 654.7 \text{ cm}^2$

$$\frac{d^2S}{dx^2} = \sqrt{3} + \frac{8000\sqrt{3}}{x^3} > 0 \text{ since } x \text{ is positive, therefore } S \text{ is minimum at } x = 4000^{\frac{1}{3}}$$



For $10 \leq x \leq 20$, max value of S occurs when $x = 10$ and the value is 779.4

From the graph of S vs x , the maximum is at the extreme end of the graph.

The range of values of x for which $S \geq 670$ is $10 \leq x \leq 13.6$ or $18.4 \leq x \leq 20$

7.

(i) He could select 50 people according to the following mutually exclusive subgroups.

	10-14 years old	15 – 20 years old	20-25 years old	25+
Male	8	14	5	1
Female	7	12	2	1

He could station himself at a busy street in the estate and interview the residents until each quota has been obtained.

(ii) To obtain a sample of 80 residents using stratified sampling, he could draw random samples from the age-groups with sample size in the same proportion as the size of each age-group, using a list of the names of residents provided by the town council or residents committee of the estate.

Age-Group	Less than 15 years old	15 and 25 years old	26 years old and above
Sample Size	$\frac{3}{8} \times 80$ (= 30)	$\frac{2}{8} \times 80$ (= 20)	$\frac{3}{8} \times 80$ (= 30)

From each stratum, he could then conduct simple random sampling to obtain random samples.

8.

(i)

$$P(B|A) = 0.5$$

$$\frac{P(B \cap A)}{P(A)} = 0.5$$

$$P(B \cap A) = 0.5 \times 0.3 = 0.15.$$

(ii)

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.3 + (1 - 0.35) - 0.15 \\ &= 0.8. \end{aligned}$$

(iii)

$$\begin{aligned} \text{required probability} &= 0.8 - 0.15 \\ &= 0.65. \end{aligned}$$

9

(i) Let X be the number of days it will rain, out of 7 days. Then, $X \sim B(7, 0.7)$.

$$\begin{aligned} P(X \geq 4) &= 1 - P(X \leq 3) \\ &= 0.87396. \end{aligned}$$

(ii)

Let Y be the number of weeks, out of 20 weeks, in which will rain at least 4 days per weeks. Then $Y \sim B(20, 0.87396)$.

$$\begin{aligned} P(9 < Y < 15) &= P(Y \leq 14) - P(Y \leq 9) \\ &= 0.0323 \end{aligned}$$

(iii)

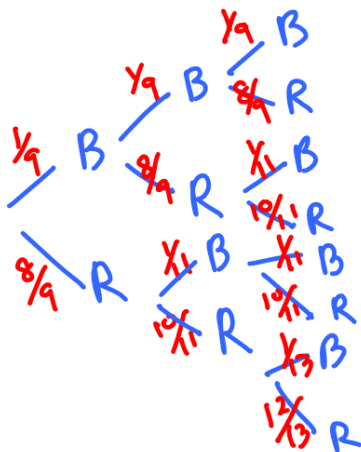
Let W be the number of days it will rain, in 20 weeks. Then $W \sim B(140, 0.7)$.

Since $n = 140$ is large, $np = 140(0.7) = 98 > 5$ and $np = 140(0.3) = 42 > 5$, $W \sim N(98, 29.4)$ approximately.

$$\begin{aligned} P(W \leq 100) &\rightarrow P(W \leq 100.5) \\ &= 0.678 \end{aligned}$$

Mean is 98 and variance is 29.4.

10.



$$\begin{aligned} \text{(i) } P(\text{consolation prize}) &= P(BBR) + P(BRB) + P(RBB) \\ &= \left(\frac{1}{9}\right)\left(\frac{1}{9}\right)\left(\frac{8}{9}\right) + \left(\frac{1}{9}\right)\left(\frac{8}{9}\right)\left(\frac{1}{11}\right) + \left(\frac{8}{9}\right)\left(\frac{1}{11}\right)\left(\frac{1}{11}\right) \\ &= 0.0273. \end{aligned}$$

(ii) $P(\text{consolation prize} \mid \text{second draw is blue})$

$$= \frac{P(\text{consolation prize} \cap \text{second draw is blue})}{P(\text{second draw is blue})}$$

$$= \frac{\left(\frac{8}{9}\right)\left(\frac{1}{11}\right)\left(\frac{1}{11}\right) + \left(\frac{1}{9}\right)\left(\frac{1}{9}\right)\left(\frac{8}{9}\right)}{\left(\frac{1}{9}\right)\left(\frac{1}{9}\right) + \left(\frac{8}{9}\right)\left(\frac{1}{11}\right)}$$

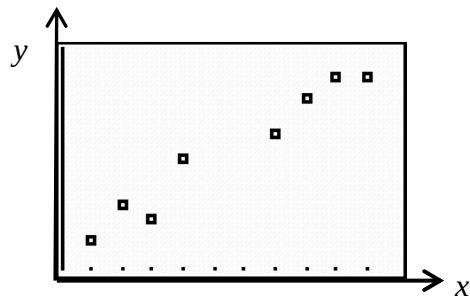
$$= 0.197.$$

$$P(\text{grand prize}) = \left(\frac{1}{9}\right)^3 = \frac{1}{729}$$

$$P(\text{wins grand prize twice}) = 3 \times \left(\frac{1}{729}\right)^2 \times \left(1 - \frac{1}{729}\right) = 5.64 \times 10^{-6}$$

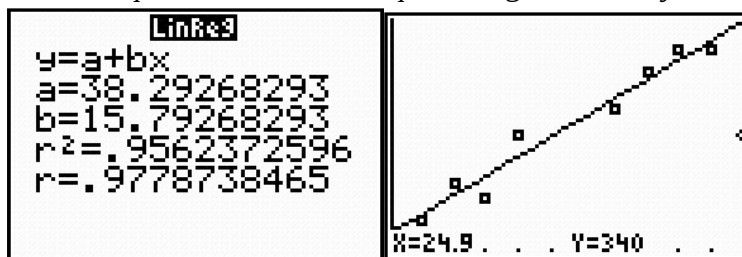
11.

(i)



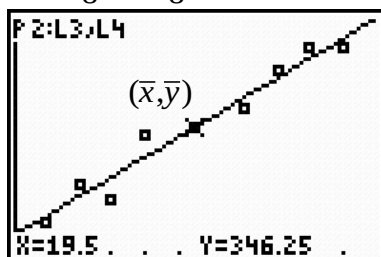
(ii)

By G.C., the equation of the least squares regression of y on x is $y = 15.7927x + 38.2927$.



(iii)

$$\bar{x} = \frac{\sum x}{8} = \frac{156}{8} = 19.5 \text{ and } \bar{y} = \frac{\sum y}{8} = \frac{2770}{8} = 346.25.$$



(iv) By G.C., $r \approx 0.978$.

Since $r \approx 0.978$, which is close to 1 and the scatter diagram suggests a linear relationship, between y and x . Thus there is a strong positive linear correlation between the number of sport car sold per quarter and the quarterly profits.

(v) When $x = 20$, $y = 354.146$, therefore \$345146. (If use regression line in part (ii), the answer would be \$345147). Both are acceptable.

Since r is close to 1 and 20 is within the data range of x , therefore the estimate is reliable.

12.

Unbiased estimate for the population mean, $\bar{x} = \frac{\sum(x-80)}{200} + 80$

$$= \frac{746.5}{200} + 80$$
$$= 83.7325 = 83.7 \text{ (3 sig. fig.)}$$

Unbiased estimate for the population variance

$$s^2 = \frac{1}{199} \left[\sum(x-80)^2 - \frac{(\sum(x-80))^2}{200} \right]$$
$$= \frac{1}{199} \left[22178.9 - \frac{746.5^2}{200} \right]$$
$$= 97.45$$
$$= 97.5 \text{ (3 sig. fig.)}$$

(ii) Let μ denote the mean retention time in hours for the population.

To test $H_0 : \mu = 83$

$H_1 : \mu \neq 83$

Significance level: 5%

Test Statistic:

Under H_0 , $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim N(0,1)$ approx'ly by central limit theorem,

Rejection criteria: reject H_0 if $p\text{-value} < 0.05$

$\bar{x} = 83.7325$, $n = 200$, $s = \sqrt{97.45019} = 9.8716865$

Using GC, $p\text{-value} = 0.2940049$

Since $p\text{-value} > 0.05$, there is insufficient evidence to reject H_0 at 5% significance level and there is insufficient evidence to say that the mean retention time is not 83 hours.

(iii) 'At the 5% significance level' means that there is a probability of 0.05 of wrongly concluding that mean retention time is not 83 hrs when it is in fact 83 hours.

(iv) To test $H_0 : \mu = 83$

$$H_1 : \mu \neq 83$$

Significance level: 5%

Test Statistics: Under H_0 , $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ approx. by CLT

$$\bar{X} \sim N\left(83, \frac{11^2}{200}\right) \text{ approximately by CLT}$$

$$n = 200, \sigma = 11$$

Since H_0 is rejected, $P(\bar{X} \leq \bar{x}) < \frac{0.05}{2}$ or $P(\bar{X} \geq \bar{x}) < \frac{0.05}{2}$

$$P(\bar{X} \leq \bar{x}) < 0.025 \text{ or } P(\bar{X} \geq \bar{x}) > 0.975$$

$$\bar{x} < 81.476 \text{ or } \bar{x} > 84.524$$

Hence range of values of $\bar{x}$ is $\bar{x} < 81.5$ or $\bar{x} > 84.5$ (3 sig. fig.)

There is no need to assume the distribution of X since the sample size = 200 is large. Therefore Central Limit Theorem ensures that the distribution of sample mean, $\bar{X}$ is normal.

13.

Let W and T denote the waiting time for the bus and travelling time on the bus respectively.

(i) $W \sim N(11, 3^2)$

$$P(W \leq 5) = 0.02275$$

$$\text{Required probability} = P(W \leq 5). P(W > 5).2 = 0.0445$$

(ii)(a) $W \sim N(11, 3^2), T \sim N(20, 5^2)$

Let X denote the total time she takes to go to work on a morning.

$$X = W + T + 2$$

$$X \sim N(11 + 20 + 2, 3^2 + 5^2) \quad \text{i.e. } X \sim N(33, 34)$$

$$P(35 < X < 40) = 0.2508 = 0.251 \text{ (3 sig. fig.)}$$

(ii)(b) $P(X > k) \leq 0.8$

$$P(X \leq k) \geq 0.2$$

$$\text{Using GC, } k \geq 28.0925$$

Hence smallest k is 29 minutes (nearest minute)

Let Y denote the total time Ann spent on travelling to and from work daily.

$$Y = X + \frac{3}{2} X = \frac{5}{2} X$$

$$Y \sim N(82.5, 212.5)$$

$$P(Y < 60) = 0.0614 \text{ (3 sig. fig.)}$$